

Alessandro Demo

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EDUCATION

Polytechnic of Turin

Master's Degree in Computer Engineering, Artificial Intelligence and Data Analytics

Turin, Italy

Nov. 2025 – Present

Relevant Courses:

- * Data Science and Database Technology
- * Big Data Processing and Analytics
- * Advanced Machine Learning

University of Turin

Bachelor's Degree in Computer Science

Turin, Italy

Sep. 2022 – Nov. 2025

Relevant Courses:

- * Data Structures and Algorithms
- * Human-Computer Interaction and Web Technologies
- * Development of Software Applications

EXPERIENCE

Technical Designer Intern

Interactive Fully Electrical Vehicles (I-FEVS)

July 2021

La Loggia, Italy

- Designed the electronic architecture for an e-bike prototype, ensuring reliable embedded system performance.
- Programmed and tested the Portenta H7 board using MicroPython to validate core hardware functionalities.
- Authored comprehensive and precise technical documentation detailing the embedded hardware components.

PROJECTS

Geospatial Data Benchmarking System | *Python, GeoPandas, DuckDB, PostgreSQL* May 2025 – Aug. 2025

- Conducted a comparative performance analysis of DuckDB, PostGIS, and GeoPandas for geospatial data processing.
- Designed and implemented benchmarking methodologies in Python to evaluate and document query execution times.
- Managed and processed complex spatial databases utilizing both vector and raster data structures.
- Source code: github.com/dimuzzo/testing-project

Sinner Tracker Dashboard | *Python, GitHub Actions, RapidAPI, Chart.js*

Apr. 2026 – Present

- Engineered a high-performance Progressive Web App (PWA) to track real-time ATP tennis statistics and match data.
- Built automated daily data ingestion pipelines using GitHub Actions to fetch and process data via RapidAPI.
- Visualized complex Head-to-Head (H2H) statistics and ranking progression using interactive charts and responsive UI design.
- Source code: github.com/dimuzzo/sinner-tracker

Quiz App Engine | *JavaScript, HTML/CSS*

Jun. 2025 – Apr. 2026

- Developed a responsive, state-managed interactive application using vanilla JavaScript and DOM manipulation.
- Built the core logic and user interface without frontend frameworks to solidify fundamental programming and web proficiencies.
- Source code: github.com/dimuzzo/quiz-app

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, R, JavaScript, HTML/CSS

Data Science & AI: Pandas, NumPy, GeoPandas, Jupyter Notebook

Databases: PostgreSQL, PostGIS, DuckDB

Tools & Infrastructure: Git, GitHub Actions, Linux/Unix, Makefiles, Arduino

Spoken Languages: Italian (Native), English (B2 First Certificate)